

Door Security Alarm System

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Abstract

This project presents the comprehensive design and implementation of an automatic door security alarm system that activates upon the unauthorized opening of a door. Engineered to provide a fundamental, localized safety alert, the system utilizes a magnetic sensor (Reed Switch) paired with a visual (LED) and auditory (piezoelectric buzzer) indicator. The core functionality relies on solid-state electronic logic: when the door opens, a disrupted magnetic field alters the switch state, completing an electrical circuit that triggers the alarm. This report details the theoretical principles of magnetic sensing, electronic circuit design, component calculations, and potential future integrations into broader smart-home ecosystems.

1. Introduction

1.1 Problem Statement

In contemporary society, localized security systems are paramount for safeguarding residential, commercial, and personal spaces. Many existing security systems are overly complex, expensive, or require continuous network connectivity. There is a distinct need for an autonomous, low-cost, and reliable alarm system that functions offline and without human intervention.

1.2 Project Objective

This project aims to design a robust, cost-effective door alarm system using basic electronic control principles. By utilizing a Reed Switch sensor to detect the precise moment a physical threshold (a door) is breached, the system immediately alerts nearby individuals via sound and light. This foundational concept is highly adaptable for various applications, including securing lockers, monitoring restricted areas, and serving as a sub-module in larger security networks.

2. Scientific Principle and Operating Mechanism

2.1 Magnetic Sensing and Ferromagnetism

The foundational principle of this system is magnetic sensing, facilitated by a **Reed Switch**.

- A Reed Switch consists of a hermetically sealed glass envelope filled with an inert gas (to prevent oxidation). Inside are two ferromagnetic, flexible metal contacts (reeds).

- When an external magnetic field is introduced, the reeds become magnetized. The opposite poles attract each other, causing the reeds to flex and touch, thereby completing an electrical path.

2.2 System Logic and State Changes

In this project, the system logic is inverted using a specific circuit design to act as a door alarm. We monitor the *absence* of the magnetic field to trigger the alarm.

The operational states are defined as follows:

1.  **State 0 (Door Closed - Standby):** The permanent magnet (attached to the door) is in close proximity to the Reed Switch (attached to the frame). The magnetic field keeps the reed switch closed. The circuit logic routes current away from the alarm. → **Alarm OFF**
2.  **State 1 (Door Open - Triggered):** The door swings open, removing the magnet from the sensor's vicinity. The magnetic field dissipates, causing the reeds to separate (switch opens). This break in the sensor path redirects current through an amplifying transistor, closing the main power circuit to the output devices. → **Alarm ON**

3. Components and Specifications

The system requires carefully selected electronic components to ensure reliability and power efficiency.

Component	Specification / Details	Function & Placement
Reed Switch	Normally Open (NO) glass tube switch.	Acts as the primary sensor. Mounted on the stationary door frame.
Permanent Magnet	Neodymium or strong ceramic magnet.	Generates the actuating magnetic field. Mounted on the moving door.
Power Source	9V DC Battery.	Provides isolated, continuous electrical power to drive the circuit.
NPN Transistor	BC547 or 2N2222 (General Purpose).	Acts as an electronic switch. Amplifies the small control current to drive the higher-load buzzer.
Buzzer	5V - 9V Piezoelectric Buzzer.	Serves as the primary auditory output device.
LED	5mm Standard Red LED.	Serves as the visual output device.
Resistors	10k Ω (Base), 220 Ω or 330 Ω (LED).	Controls current flow to protect sensitive components (Transistor base and LED).

4. In-Depth Circuit Design

The circuit is designed using an NPN transistor configured as a solid-state switch.

4.1 Wiring Configuration

- **The Pull-Down Logic:** The Reed Switch is connected between the base of the NPN transistor and the ground (negative terminal). A $10\text{k}\Omega$ resistor connects the positive terminal to the transistor's base.
- **When the Door is Closed:** The magnet keeps the Reed Switch closed. This creates a path of least resistance directly to the ground. The current bypasses the transistor's base, keeping the transistor **OFF**. The buzzer and LED, connected to the transistor's collector, remain unpowered.
- **When the Door Opens:** The magnet moves away, and the Reed Switch opens. The path to ground is broken. Current is now forced to flow through the $10\text{k}\Omega$ resistor into the **Base** of the NPN transistor. This base current turns the transistor **ON**, allowing a much larger current to flow from the Collector to the Emitter, powering the Buzzer and LED.

5. Mathematical Calculations

To ensure the longevity of the components, proper resistance values must be calculated using Ohm's Law ($V = I \times R$).

5.1 LED Current Limiting Resistor

If we use a 9V battery, applying 9V directly to a standard LED will cause it to burn out immediately. We must calculate the required resistor.

- Source Voltage (V_s) = 9V
- LED Forward Voltage (V_f) $\approx 2\text{V}$ (Typical for a Red LED)
- Desired LED Current (I) $\approx 20\text{mA} = 0.02\text{A}$

Using the formula for the limiting resistor (R):

$$R = \frac{V_s - V_f}{I}$$

$$R = \frac{9\text{V} - 2\text{V}}{0.02\text{A}}$$

$$R = \frac{7\text{V}}{0.02\text{A}} = 350\Omega$$

Conclusion: A standard 330Ω or 220Ω resistor is sufficient to protect the LED while maintaining adequate brightness.

5.2 Transistor Base Resistor

To ensure the transistor reaches full saturation (turns fully ON) without drawing excessive current from the battery when the alarm triggers, a $10\text{k}\Omega$ resistor is standard. Current at the base (I_b) when the switch opens:

$$I_b = \frac{V_s - V_{be}}{R_{base}}$$

$$I_b = \frac{9V - 0.7V}{10,000\Omega} = 0.83mA$$

This small base current is perfectly sufficient to allow the transistor to switch on the heavier load of the buzzer (which typically requires 20mA to 50mA).

6. Implementation and Assembly

1. **Breadboard Prototyping:** The circuit was first assembled on a breadboard to verify the transistor logic and resistor calculations.
2. **Soldering & Housing:** Components were soldered onto a perforated PCB for durability. The circuit was housed in a small, discrete plastic enclosure to protect the wiring.
3. **Installation:** The magnetic unit was adhered to the moving edge of the door using heavy-duty double-sided tape. The main enclosure (containing the sensor, circuit, and buzzer) was aligned precisely on the door frame, ensuring a gap of less than 1 cm when the door is closed.

7. Future Works and Improvements

While the current system provides robust localized functionality, several enhancements can be integrated for aerospace or advanced smart-home applications:

- **IoT and Wireless Connectivity:** Integration with an ESP8266 or ESP32 microcontroller to send wireless alerts (via Wi-Fi or Bluetooth/RF modules) to a smartphone or a central security hub.
- **Multi-Point Detection:** Incorporating multiple reed switches in a parallel matrix to monitor several doors, windows, or equipment hatches simultaneously.
- **Sensitivity Optimization:** Upgrading the magnet to a higher-grade Neodymium (e.g., N52) to achieve higher sensitivity and allow for a wider operational gap between the door and frame.
- **Power Efficiency Upgrades:** Implementing a "sleep mode" microcontroller setup or solar-assisted battery charging, making the system suitable for long-term, remote deployments.

8. Conclusion

The Door Security Alarm System project successfully demonstrates the design and operation of a reliable, automatic security mechanism. By leveraging the principles of magnetic fields and solid-state transistor switching, the team developed an accessible, low-cost, and highly practical electronic device. The foundational concept established here provides a robust platform that meets basic security needs while remaining highly adaptable for future technological expansions.